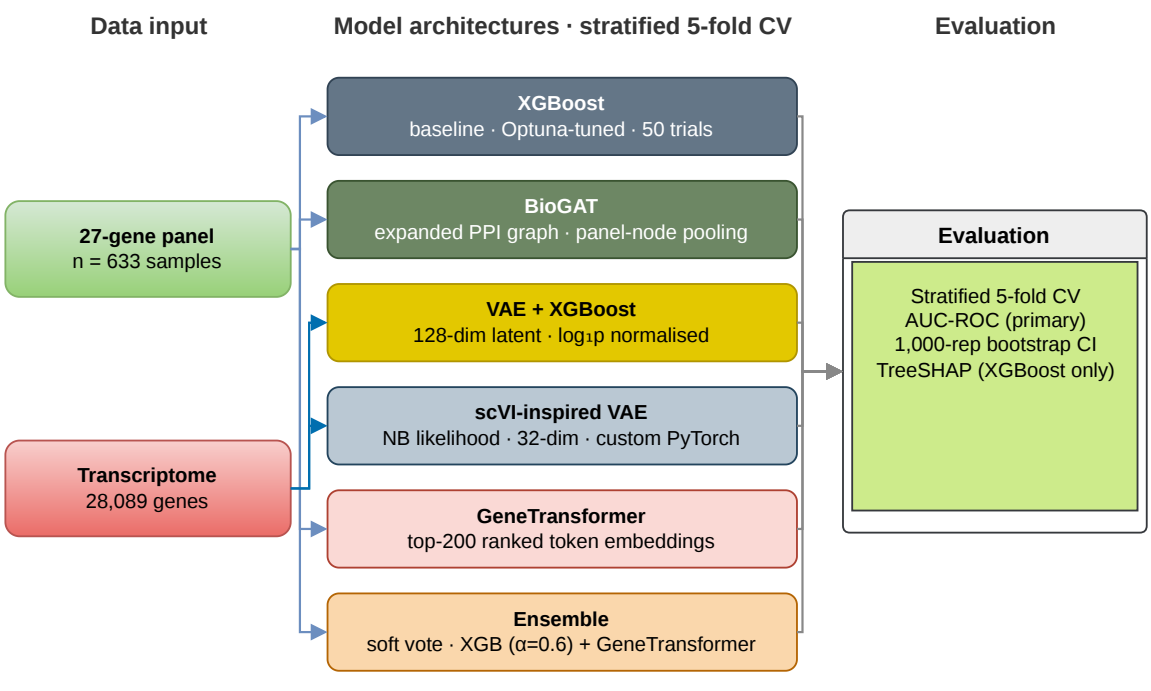

Data input
Model architectures · stratified 5-fold CV
Evaluation
27-gene panel
n = 633 samples
Transcriptome
28,089 genes
XGBoost
baseline · Optuna-tuned · 50 trials
BioGAT
expanded PPI graph · panel-node pooling
VAE + XGBoost
128-dim latent · log₁p normalised
scVI-inspired VAE
NB likelihood · 32-dim · custom PyTorch
GeneTransformer
top-200 ranked token embeddings
Ensemble
soft vote · XGB (α=0.6) + GeneTransformer
Evaluation
Stratified 5-fold CV
AUC-ROC (primary)
1,000-rep bootstrap CI
TreeSHAP (XGBoost only)